

Pardeep Kumar

SCIENTIFIC COMPUTING RESEARCHER | NUMERICAL METHODS | EDUCATION & OUTREACH

Amsterdam, the Netherlands | +31 6 1311 9813 | pardeep.iitb@gmail.com | [Portfolio](#)
[LinkedIn](#) | [Google Scholar](#) | Languages: English, Hindi, Dutch (A2)

PROFILE

Scientific computing researcher with a submitted doctoral thesis developed at [CWI Amsterdam](#) and [TU Delft](#), and more than ten years of experience in numerical modelling, simulation software, and computational problem-solving. My research concerns robust numerical methods for multiphase thermodynamics and CO₂-rich transport.

My interest in high-school education centres on the parts of research I value most: asking careful questions, making difficult ideas understandable, and helping others reason independently. My background supports clear explanations and computational examples in mathematics, physics, and computing education.

EDUCATION AND OUTREACH

My education interests focus on helping high-school students approach mathematics and physics through careful questions, physical intuition, visual explanation, and small computational investigations.

- Topics include high-school mathematics, physics, computational thinking, and science outreach.
- Able to connect classroom ideas to real examples from fluids, thermodynamics, electromagnetics, aerospace, and engineering.
- Committed to treating questions and mistakes as useful parts of learning, while helping students learn how to test an argument.

EDUCATION

PhD in Mechanical Engineering

2022-2026

Delft University of Technology | Research conducted at [CWI Amsterdam](#)

Thesis submitted; defence planned for January 2027. Research focus: numerical methods for transient multiphase flow of multicomponent mixtures, with industrial collaboration from Shell Projects & Technology.

MSc in Aerospace Engineering

2012-2014

Indian Institute of Technology Bombay, India

Research focus: finite-volume time-domain methods for electromagnetic propagation and scattering.

DOCTORAL RESEARCH AND SCIENTIFIC CONTRIBUTION

- Reformulated constrained phase-equilibrium calculations in a more efficient thermodynamic variable space.
- Coupled phase-stability and equilibrium calculations to finite-volume solvers for CO₂-rich tank and pipeline transport.
- Developed thermodynamically consistent temperature-evolution models for multiphase flow.
- Investigated exact and approximate Riemann problems for real-fluid mixtures.
- Created Julia and Python research software supported by systematic validation and reproducibility workflows.

TRANSFERABLE STRENGTHS FOR EDUCATION

Clear explanation Used to moving between mathematics, physical meaning, algorithms, plots, and plain-language interpretation.	Interdisciplinary perspective Examples drawn from aerospace, energy, fluids, electromagnetics, semiconductors, finance, and computing.
Evidence and verification A research habit of examining assumptions, limiting cases, numerical results, and alternative explanations.	Computational demonstrations Able to create small programs and visual models that let students investigate a mathematical or physical question.

SELECTED PROFESSIONAL DEVELOPMENT AT TU DELFT

Graduate School courses: Scientific Storytelling; Advanced Problem Solving and Decision Making; Teamwork, Leadership and Group Dynamics; Conversation Skills; Embedding Societal Values in Research; and Scientific Integrity.

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SCIENTIFIC COMPUTING AND ENGINEERING EXPERIENCE

SCIENTIFIC COMPUTING STRENGTHS

Numerical methods Finite-volume and finite-difference methods, Riemann solvers, nonlinear equations, constrained optimisation, automatic differentiation, and numerical linear algebra.	Computational physics Multiphase flow, real-fluid thermodynamics, CFD, electromagnetics, wave propagation, and numerical relativity.
Scientific programming C++, Julia, Python, C#, F#, Java, MATLAB, CUDA, MPI, multithreading, and performance-oriented computation.	Research software practice Modular architecture, interoperability, testing, validation, version control, visualisation, and reproducible computation.

PROFESSIONAL EXPERIENCE

PhD Researcher in Scientific Computing

Aug 2022-Aug 14, 2026

Centrum Wiskunde & Informatica (CWI), Amsterdam

- Developed numerical formulations and software for multicomponent, multiphase thermodynamics and transient flow.
- Connected mathematical analysis, physical modelling, algorithm design, implementation, and validation in interdisciplinary research.

Software Engineer

Feb-Aug 2022

ASM International, Almere

- Enhanced and refactored a Java-based simulator for semiconductor thin-film deposition equipment.

Software Engineer

Jun 2019-Dec 2021

Shell, Amsterdam | Individual contributor

- Migrated numerical workflows from Python to C++ and CUDA, reducing a representative runtime from about 30 minutes to 15 seconds.
- Developed concurrent infrastructure, interoperability layers, and visualisation for process and pipeline simulation software.

Software Engineer

Feb-Apr 2019

Aakraya Research, Bangalore | Team of 2

- Developed regression-model generation and live profit-and-loss evaluation tools for high-frequency trading.
- Improved C++ order-book infrastructure for efficient order matching.

Software Engineer II

May-Nov 2018

KLA-Tencor, Chennai | Team of 2

- Developed cross-application infrastructure and a silica-wafer thickness simulation engine using C#, F#, and MATLAB.
- Implemented reusable testing libraries and an Erlang fault-tolerance mechanism with failover and recovery.

Software Engineer

Mar 2016-Apr 2018

Altair, Bangalore | Team of 6

- Developed shared-memory infrastructure and geometry and mesh-manipulation tools in C++.
- Modernised C++/Python bindings, built Python testing frameworks, and contributed to an MVC refactoring.

Research and Development Engineer

Jul 2014-Mar 2016

Fluidyn, Bangalore | Team of 4

- Developed an MPI-parallel finite-volume electromagnetic solver with higher-order spatial schemes.
- Implemented near-to-far-field transformations and extended a Navier-Stokes solver for aeroacoustic prediction.

SELECTED PEER-REVIEWED PUBLICATIONS

1. P. Kumar and P. I. Rosen Esquivel. A Reformulation of UVN-Flash for Multicomponent Two-Phase Systems with Application to CO₂-Rich Mixture Transport in Pipelines. *Computers & Fluids*, 314, 107108, 2026.
2. P. Kumar and P. I. Rosen Esquivel. Solving the UVN-Flash Problem in TVN-Space. *Fluid Phase Equilibria*, 599, 114528, 2026.
3. P. Kumar, B. Sanderse, P. I. Rosen Esquivel, and R. A. W. M. Henkes. A New Temperature Evolution Equation That Enforces Thermodynamic Vapour-Liquid Equilibrium in Multiphase Flows. *Computers & Fluids*, 289, 106524, 2025.

WEB APPLICATIONS

I independently developed The Republic, Nazar India, Chess Duel, and WorkAtlas. The work demonstrates product thinking, full-stack delivery, and self-directed learning alongside my scientific-computing and education interests. [View web application experience](#)